
Weak-to-Strong Generalization via Direct On-Policy Distillation

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Abstract

Reinforcement learning with verifiable rewards (RLVR) significantly improves language model reasoning. However, its reliance on sparse terminal rewards makes **credit assignment** over long chains of thought fundamentally difficult; to compensate, standard RL requires extensive rollout sampling, which becomes computationally prohibitive as model sizes scale. Yet a smaller model that has already undergone RL has **implicitly discovered exactly this elusive credit-assignment signal**. The central question is how to efficiently transfer this localized knowledge to a larger target model. We propose **Direct On-Policy Distillation (Direct-OPD)**, a novel **weak-to-strong generalization** paradigm that transfers the small teacher’s RL-induced **policy shift** rather than its absolute endpoint distribution. By applying the policy-as-reward identity in reverse, Direct-OPD treats the log-ratio between the post-RL teacher and its pre-RL reference as a **dense, token-level implicit reward**. This formulation allows the student to be optimized directly on its own on-policy states, bypassing explicit reward modeling and the prohibitive exploration costs of target-model RL. Empirical evaluations show that Direct-OPD consistently leverages weaker teachers to improve stronger target models; notably, it boosts Qwen3-1.7B from 48.3% to 62.4% on AIME 2024 in just 4 hours on 8 A100 GPUs. It outperforms step-matched direct RL and enables the sequential composition of multiple policy shifts. Our results demonstrate that reinforcement learning outcomes can be efficiently reused across model scales as directional improvement signals.

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1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has become a dominant post-training recipe for improving the reasoning ability of large language models [1, 2]. Its reward, however, is sparse, grading an entire chain of thought by a single outcome at the end. Such thin supervision yields a high-variance learning signal, and RL relies on heavy sampling—many rollouts and many update steps—to average out the noise. Training is therefore slow, and it becomes slower still on larger models, where generating each rollout is more expensive. At root, this expense reflects a credit-assignment problem: a single terminal reward gives no direct indication of which of the thousands of intermediate tokens deserved it.

In practice, RLVR is run most often with one of two policy-gradient algorithms. GRPO [1, 3] avoids a value model: for each prompt it samples a group of responses, scores them with the verifiable reward, and normalizes the scores within the group to form an advantage, which it assigns to every token of a response. PPO [4] is instead an actor-critic method: it trains a value model to estimate the return at each state and combines it with the verifiable reward to form token-level advantages, then updates the policy under a clipped objective. Both learn from the same sparse terminal reward and both rely on generating large numbers of rollouts, so neither escapes the basic cost: running RLVR on each new and larger model repeats this rollout loop from scratch. This motivates the central question of our work: if a smaller model has already been through RL, can we reuse the resulting model to improve a larger one, instead of re-running sparse-reward RL at the large scale?

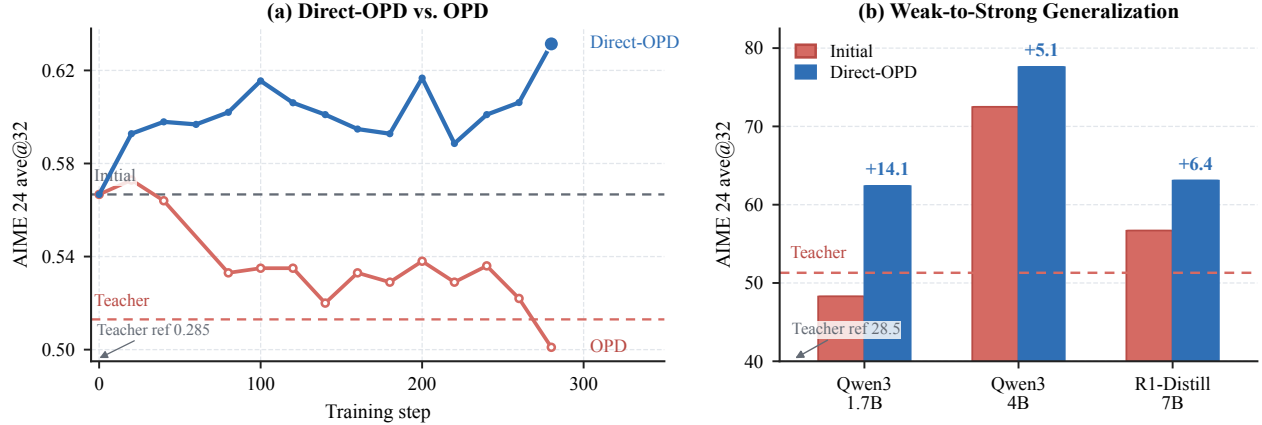


Figure 1 Direct-OPD transfers the teacher’s RL improvement rather than imitating the teacher. (a) With R1-Distill-7B as the starting student and JustRL-1.5B—obtained by running RL on R1-Distill-1.5B—as the teacher, vanilla OPD directly degrades the student below the teacher, while transferring the JustRL-1.5B – R1-Distill-1.5B policy shift with Direct-OPD improves it instead. (b) The same policy shift improves three students on AIME 2024, including Qwen3-4B and R1-Distill-7B, whose initial accuracy already exceeds the post-RL teacher.

Reusing this trained model, however, is not straightforward. The natural tool is on-policy distillation (OPD), where the student samples its own trajectories and is pushed toward the teacher at each visited token using the teacher–student log-ratio $\log \pi_T - \log \pi_S$. This asks the student to imitate the teacher’s endpoint policy—and here the teacher is the weaker model. Figure 1(a) shows the consequence: applied to R1-Distill-7B, which starts at 56.7, well above the teacher’s 51.3, endpoint OPD steadily drags the student down to around 50, below the teacher itself. Imitating the endpoint copies the teacher’s RL improvement and its lower absolute ability at once, and for a student that already exceeds the teacher, the pull toward that lower ability outweighs the improvement.

We propose **Direct On-Policy Distillation (Direct-OPD)**, which keeps the dense, on-policy interface of OPD but changes what it subtracts. Instead of subtracting the student, Direct-OPD subtracts the teacher’s own pre-RL reference:

$$\Delta_T(y | x) = \log \pi_T(y | x) - \log \pi_{T_{\text{ref}}}(y | x), \quad (1)$$

which isolates only what RL changed in the teacher and discards its absolute ability. Under the KL-regularized RL framework, this log-ratio acts as an implicit reward: it mirrors the underlying reward landscape by scoring how much more (or less) likely a response becomes after optimization [5], see Sec. 2.3. This is because the post-RL policy π_T is the direct outcome of the teacher’s reinforcement learning, and this implicit reward natively encodes the granular credit assignment discovered on the smaller model—extracted seamlessly by comparing two static checkpoints. We borrow the term direct from Direct Preference Optimization (DPO): just as DPO fits a policy directly from preferences without a separately trained reward model, Direct-OPD uses the same policy-as-reward identity in reverse to train the student directly from this implicit reward, without running verifiable-reward RL on the target model. Concretely, the student samples its own trajectories, and at each visited prefix we evaluate Δ_T on the student’s top- k candidate tokens, encouraging tokens whose probability RL raised and discouraging those it lowered, while a KL anchor keeps the student near its initialization. Because the supervision is the teacher’s change rather than its endpoint, a weaker teacher can still improve a stronger student.

Figure 1(b) previews the payoff. The teacher pair here is R1-Distill-1.5B and JustRL-1.5B, where the latter is obtained by running RL on R1-Distill-1.5B; transferring their policy shift improves all three of Qwen3-1.7B, Qwen3-4B, and R1-Distill-7B—including Qwen3-4B and R1-Distill-7B, which already start above the post-RL teacher. The transfer is also cheap: with 8 A100 GPUs for about 4 hours, Direct-OPD raises Qwen3-1.7B from 48.3 to 62.4 on AIME 2024, comparable to what Polaris reaches by running RL directly on Qwen3-1.7B with 32 A100 GPUs for at least a week [6].

Contributions.

- **A credit-assignment view of weak-to-strong transfer.** We recast reusing RL across model scales as transferring credit assignment: a small teacher’s RL-induced policy shift serves as a dense implicit reward, which Direct-OPD applies on the student’s own on-policy states—the DPO policy-as-reward identity used in reverse.
- **Weak-to-strong gains across teacher pairs and students.** With two teacher pairs (JustRL and QuestA) and three students from two families (R1-Distill-7B, Qwen3-1.7B, and Qwen3-4B), Direct-OPD improves every student—including R1-Distill-7B and Qwen3-4B, which already start above the post-RL teacher—and lifts Qwen3-1.7B from 48.3 to 62.4 on AIME 2024 in about 4 hours on 8 A100 GPUs. We compose two policy shifts sequentially to train Qwen3-1.7B, reaching 63.8 on AIME 2024.
- **Cheaper than direct large-scale RL.** At matched RL steps, running RL on a 1.5B model and transferring with Direct-OPD beats running RL directly on R1-Distill-7B in both accuracy and compute.
- **Analysis of when the signal is reliable.** Unlike endpoint OPD, Direct-OPD does not require high teacher–student top- k overlap and transfers across thinking patterns; we identify the response-length and KL conditions under which the implicit reward stays aligned with validation accuracy.

2 Direct On-Policy Distillation

2.1 Motivation

The premise of this work is that the credit-assignment signal discovered by reinforcement learning on a smaller model can be efficiently reused to improve a stronger target student. However, standard on-policy distillation (OPD) struggles in this weak-to-strong setting. Because OPD forces the student to match the teacher’s endpoint distribution, it transfers the teacher’s capacity bottlenecks alongside its RL gains. If the student already possesses stronger foundational capabilities than the small teacher, imitating this absolute endpoint actively degrades the student’s pre-existing knowledge, pulling its performance down to the weaker teacher’s ceiling (as illustrated in Figure 1).

To unlock effective weak-to-strong generalization, we must separate the teacher’s RL-acquired reasoning improvements from its inherent capacity limits. Direct-OPD achieves this by shifting the objective from endpoint imitation to directional transfer. By isolating the RL-induced trajectory change—defined as the density ratio shift from the teacher’s pre-RL reference to its post-RL checkpoint—Direct-OPD extracts a pure, directional improvement signal. This formulation retains the dense, on-policy advantages of OPD while ensuring the student is guided exclusively by the teacher’s optimization trajectory rather than constrained by its absolute performance level.

2.2 Preliminaries: On-Policy Distillation

We consider autoregressive language-model policies. Let $x \sim \mathcal{D}$ be a prompt and $y = (y_1, \dots, y_T)$ a response, so that a policy π factorizes as $\log \pi(y | x) = \sum_{t=1}^T \log \pi(y_t | s_t)$ over prefixes $s_t = (x, y_{<t})$. We distinguish four policies: the policy being trained π_θ and its initial checkpoint π_S ; and a teacher pair consisting of a pre-RL reference $\pi_{T_{\text{ref}}}$ and the corresponding post-RL teacher π_T . The weak-to-strong setting of interest is when π_T may be smaller or weaker than the target student, yet the change from $\pi_{T_{\text{ref}}}$ to π_T still carries useful information.

On-policy distillation (OPD) supervises the student on trajectories it samples itself. Given a prompt x , the student draws $\hat{y} \sim \pi_\theta(\cdot | x)$, and at each visited prefix $s_t = (x, \hat{y}_{<t})$ we read off the student and teacher next-token distributions $p_t(v) = \pi_\theta(v | s_t)$ and $q_t(v) = \pi_T(v | s_t)$. Standard OPD minimizes the sequence-level KL to the teacher on these rollouts,

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi_T(\cdot | x))], \quad (2)$$

which factorizes exactly into dense, per-token supervision on the states the student actually visits,

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \hat{y} \sim \pi_\theta} \left[\sum_{t=1}^T D_{\text{KL}}(p_t \parallel q_t) \right]. \quad (3)$$

In practice the teacher is queried only on the student’s top- k support $S_t = \text{TopK}_v p_t$, giving the local estimator

$$\mathcal{L}_{\text{OPD}}^{\text{top-}k}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \hat{y} \sim \pi_\theta} \left[\sum_{t=1}^T D_{\text{KL}}(\bar{p}_t^{S_t} \parallel \bar{q}_t^{S_t}) \right], \quad (4)$$

where $\bar{p}_t^{S_t}$ and $\bar{q}_t^{S_t}$ renormalize p_t and q_t on S_t [7]. The per-token signal is thus the teacher–student log-ratio $\log q_t(v) - \log p_t(v)$; Direct-OPD keeps this on-policy, top- k interface but changes *what the student is compared against* (Section 2.3).

2.3 Policy Shifts as Implicit Rewards

What signal should we transfer? Direct-OPD uses the teacher’s RL-induced **policy shift**, the sequence-level log-ratio between the post-RL teacher π_T and its pre-RL reference $\pi_{T_{\text{ref}}}$,

$$\Delta_T(y \mid x) = \log \pi_T(y \mid x) - \log \pi_{T_{\text{ref}}}(y \mid x), \quad (5)$$

which isolates the teacher’s RL-induced *direction* rather than its absolute endpoint distribution. This shift is not an arbitrary heuristic: it is the teacher’s reward, recovered through the **policy-as-reward** identity from KL-regularized RL, used **in reverse**.

For a reward r , reference policy π_{ref} , and penalty $\beta > 0$, the KL-regularized objective $\max_\pi \mathbb{E}_{y \sim \pi} [r(x, y) - \beta \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)}]$ has the closed-form optimum $\pi^* \propto \pi_{\text{ref}} \exp(r/\beta)$, hence

$$\log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} = \frac{1}{\beta} r(x, y) - \log Z(x). \quad (6)$$

Since $\log Z(x)$ is constant across responses to the same prompt, the policy–reference log-ratio *is* the reward, up to a positive scale and a per-prompt constant. **DPO uses this forward:** from a reward implied by preference comparisons, it substitutes the log-ratio form of r into the preference loss and fits the policy directly, removing the explicit reward model [5]. **We use it in reverse:** we are already *given* the policy—a post-RL teacher π_T and its pre-RL reference $\pi_{T_{\text{ref}}}$, the output of the teacher’s RL—and we read the reward back out. Viewing π_T as the KL-regularized optimum of some reward r_T from $\pi_{T_{\text{ref}}}$,

$$\Delta_T(y \mid x) = \frac{1}{\beta} r_T(x, y) - \log Z_T(x), \quad (7)$$

so the shift Δ_T is exactly the teacher’s **implicit reward**, up to a positive scale and a per-prompt constant. Direct-OPD transfers this reward to the student.

2.4 The Direct-OPD Objective

The objective. With the student initialized from π_S , Direct-OPD maximizes the teacher’s implicit reward Δ_T while regularizing the student toward its own initialization,

$$J_{\text{Direct-OPD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\mathbb{E}_{y \sim \pi_\theta(\cdot|x)} [\Delta_T(y \mid x)] - \alpha D_{\text{KL}}(\pi_\theta(\cdot \mid x) \parallel \pi_S(\cdot \mid x)) \right], \quad (8)$$

where $\alpha > 0$ controls the KL penalty. This is itself a KL-regularized RL objective on the student: its optimum is

$$\pi^*(y \mid x) \propto \pi_S(y \mid x) \exp\left(\frac{1}{\alpha} \Delta_T(y \mid x)\right) = \pi_S(y \mid x) \left(\frac{\pi_T(y \mid x)}{\pi_{T_{\text{ref}}}(y \mid x)} \right)^{1/\alpha}, \quad (9)$$

and substituting the implicit-reward form of Eq. 7 gives $\pi^* \propto \pi_S \exp(r_T/\alpha\beta)$. The student is therefore optimized under KL-regularized RL whose reward is the small teacher’s implicit reward r_T and whose reference is the student’s own initialization π_S : it inherits the teacher’s RL reward directly from the checkpoint pair, never seeing a verifiable reward or running RL exploration on the target.

From sequence to token level. Because both teachers factorize autoregressively over the same prefixes, the sequence shift decomposes exactly into a sum of token-level shifts,

$$\Delta_T(y | x) = \sum_t r_t(y_t | s_t), \quad r_t(v) = \log \pi_T(v | s_t) - \log \pi_{T_{\text{ref}}}(v | s_t), \quad (10)$$

so that $r_t(v)$ is a dense per-token reward: $r_t(v) > 0$ where the teacher’s RL encouraged token v , and $r_t(v) < 0$ where it suppressed it. This lets us optimize Eq. 8 on the student’s own rollouts with token-level credit assignment, reusing the on-policy, top- k interface of Section 2.2.

Top- k action-space restriction. To keep the signal on actions the student actually considers, at each visited prefix we restrict to its top- k support $S_t = \text{TopK}_v \pi_\theta(v | s_t)$ and renormalize the student probabilities on it,

$$\bar{p}_t(v) = \frac{\pi_\theta(v | s_t)}{\sum_{u \in S_t} \pi_\theta(u | s_t)}, \quad v \in S_t, \quad (11)$$

which is the renormalized student distribution $\bar{p}_t^{S_t}$ of Section 2.2.

Analytical top- k policy gradient. The per-token reward $r_t(v)$ depends only on the teacher pair, so we use it directly as the advantage. A single-sample, token-level policy gradient for Eq. 8 would weight the log-likelihood of the sampled token by its reward,

$$\nabla_\theta J_{\text{MC}} = \mathbb{E}_{x,y} \left[\sum_t r_t(y_t) \nabla_\theta \log \pi_\theta(y_t | s_t) \right], \quad (12)$$

but estimating each step from one token is high-variance. Since we already have the full top- k rewards and probabilities at every visited prefix, we replace the single-token estimate at each step by its exact expectation over $\bar{p}_t$, i.e. we Rao–Blackwellize the per-step action while still sampling the trajectory $y \sim \pi_\theta$:

$$\nabla_\theta J_{\text{analytical}} = \mathbb{E}_{x,y \sim \pi_\theta} \left[\sum_t \sum_{v \in S_t} \underbrace{\bar{p}_t(v)}_{\text{weight}} \underbrace{r_t(v)}_{\text{reward}} \underbrace{\nabla_\theta \log \pi_\theta(v | s_t)}_{\text{log-likelihood}} \right]. \quad (13)$$

This is the exact expected per-step gradient under the top- k action distribution; it removes the token-sampling variance of $\nabla_\theta J_{\text{MC}}$ while leaving the trajectory distribution untouched.

Stop-gradient coefficient. For the policy-gradient identity to hold, the weight $\bar{p}_t(v)$ must enter only as a scalar. It depends on θ through the student softmax, so differentiating it would inject spurious Jacobian terms from the top- k distribution. We therefore detach the weighted reward into a static coefficient,

$$A_t^w(v) = \text{stop_gradient}(\bar{p}_t(v) \cdot r_t(v)), \quad (14)$$

giving the final local top- k surrogate,

$$\nabla_\theta J_{\text{Direct-OPD}} \approx \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta} \left[\sum_t \sum_{v \in S_t} A_t^w(v) \nabla_\theta \log \pi_\theta(v | s_t) \right] - \alpha \nabla_\theta D_{\text{KL}}(\pi_\theta(\cdot | x) \| \pi_S(\cdot | x)). \quad (15)$$

In practice we do not differentiate the KL analytically; we use the standard KL-penalty implementation in `verl` to anchor π_θ to π_S .

2.5 Adaptive KL Control

The reward Direct-OPD transfers carries a scale it does not get to choose. By Eq. 7, $\Delta_T = \frac{1}{\beta} r_T - \log Z_T$: its magnitude is the teacher’s reward divided by the teacher’s KL budget β , both fixed when the teacher was trained—encoded in how far that RL pushed π_T from $\pi_{T_{\text{ref}}}$ —and never observable from the checkpoint pair. The per-token shift $r_t(v) = \log \pi_T(v | s_t) - \log \pi_{T_{\text{ref}}}(v | s_t)$ is likewise a function of the teacher pair alone; the student enters only through where the signal is read off—the prefixes its rollouts visit and the top- k actions retained at each one.

This is what makes a fixed α for all settings ill-posed rather than merely hard to tune. From the optimum in Eq. 9, $\pi^* \propto \pi_S \exp(\frac{1}{\alpha} \Delta_T)$, α is the temperature of an exponential tilt of π_S , and is meaningful only relative to the scale of Δ_T . But Δ_T lives in the teacher’s reward units ($\frac{1}{\beta} r_T$) while α is a coefficient on the student’s KL—two scales with no a priori conversion, the teacher’s being unobservable. No single α can therefore be calibrated in advance; what should be held stable is not the coefficient but the balance between the dense transfer reward and the anchor to π_S . We maintain this balance with a lightweight controller that adapts α to the running scale of the signal rather than fixing it.

Let α_m be the KL coefficient at training iteration m , and let $\bar{r}_m$ be the batch-level mean of the student-weighted shift $\bar{p}_t(v) r_t(v)$ over the visited prefixes and their top- k candidates—the same dense reward the gradient optimizes. Before the actor update we set

$$\alpha_{m+1} = \text{clip}(\alpha_m (1 + \epsilon \text{sgn}(\bar{r}_m)), \alpha_{\min}, \alpha_{\max}), \quad (16)$$

with $\text{sgn}(0) = 0$; by default $\epsilon = 0.01$ and $[\alpha_{\min}, \alpha_{\max}] = [0.5, 2.5]$. Here $\text{sgn}(\bar{r}_m)$ reports whether, on the prefixes the student currently visits, the teacher’s RL on average raised or lowered the retained candidate tokens—a batch-level sign that the student-weighted shift carries directly.

The controller targets the balance point of the dense signal directly. When the visited prefixes carry positive teacher shift on average, it raises α to keep the student from over-amplifying that local signal; when the average shift is negative, it lowers α , weakening the anchor so the dense gradient can move probability away from tokens whose likelihood the teacher’s RL reduced. This is deliberately unlike the standard adaptive controller for an in-reward KL penalty, which steers toward a target KL value: here the update is driven by the sign of the Direct-OPD dense reward and acts on the explicit actor KL loss coefficient. We provide a detailed analysis of KL-coefficient sensitivity in Section 4.3.

3 Experiments

In this section we answer the following research questions:

- **RQ1.** Can a small teacher’s RL-induced policy shift improve students that already match or exceed the teacher’s own ability, and does this hold across different teacher pairs and student families? (Section 3.1)
- **RQ2.** For a fixed budget, does running RL on a small model and transferring its policy shift with Direct-OPD outperform running RL directly on the large target—in both accuracy and compute? (Section 3.2)
- **RQ3.** Can several independently learned policy shifts be composed sequentially to accumulate their gains in a single student? (Section 3.3)

3.1 A small RL teacher improves much stronger students

Our first experiment asks whether an RL improvement learned by a small teacher can be used to improve students that are much stronger. We first use R1-Distill-1.5B as the teacher reference and JustRL-1.5B as the post-RL teacher, then distill the policy shift from this pair into three target students: R1-Distill-7B, Qwen3-1.7B, and Qwen3-4B. We also test a second teacher pair, Nemotron-1.5B $\rightarrow$ QuestA-Nemotron-1.5B, which comes from a different training pipeline and data source, and transfer its policy shift into R1-Distill-7B and Qwen3-1.7B.

Figure 2 shows that the transferred policy shift improves students across these settings, including R1-Distill-7B and Qwen3-4B students whose starting accuracies already exceed the post-RL JustRL teacher. The QuestA results further show that the effect is not tied to one teacher family or one RL recipe: Direct-OPD generalizes across teacher pairs, student model families, training data, and training pipelines.

This result supports the interpretation that Direct-OPD uses an RL-induced direction of improvement rather than the teacher’s full policy.

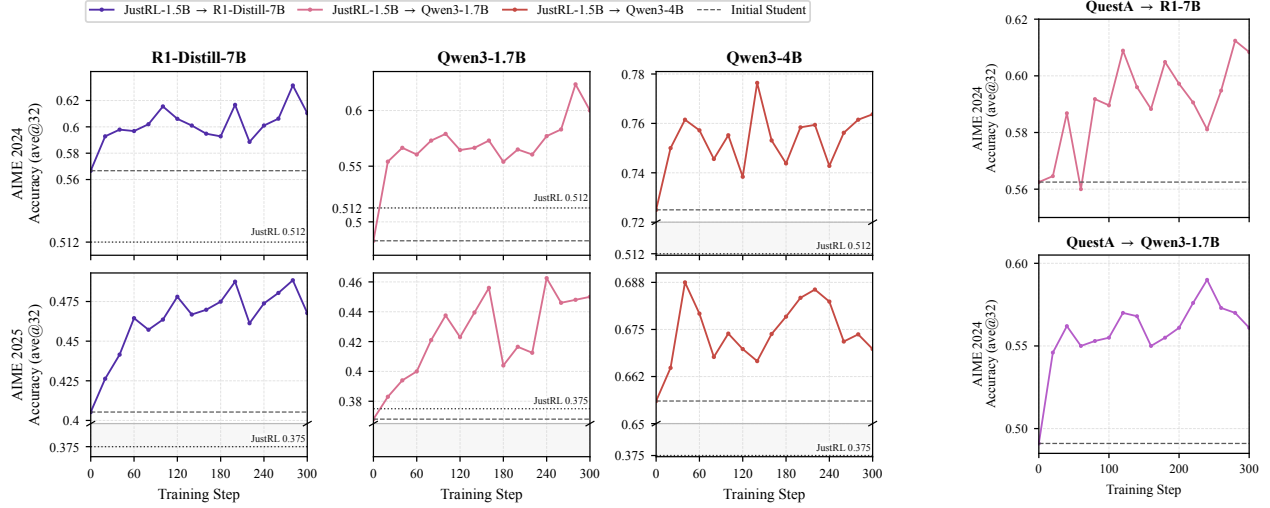


Figure 2 Direct-OPD transfers RL-induced policy shifts across teacher pairs and student families. **Left:** R1-Distill-1.5B → JustRL-1.5B transfer into R1-Distill-7B, Qwen3-1.7B, and Qwen3-4B, evaluated on AIME 2024 and AIME 2025. **Right:** Nemotron-1.5B → QuestA-Nemotron-1.5B transfer into R1-Distill-7B and Qwen3-1.7B on AIME 2024. The two teacher pairs come from different training data and pipelines, showing that Direct-OPD is not specific to a single post-RL teacher.

(a) JustRL policy-shift transfer			(b) QuestA policy-shift transfer		
Model	AIME24	AIME25	Model	AIME24	AIME25
Teacher ref	28.5	24.0	Teacher ref [8]	61.77	49.50
Teacher RL [2]	51.3	37.5	Teacher RL [8]	72.50	62.29
Qwen3-1.7B	48.3	36.8	Qwen3-1.7B	49.1	36.8
+ Direct-OPD	62.4 +14.1	46.3 +9.5	+ Direct-OPD	59.0 +9.9	43.1 +6.3
Qwen3-4B	72.5	65.6	R1-Distill-7B	56.3	39.5
+ Direct-OPD	77.6 +5.1	68.8 +3.2	+ Direct-OPD	61.2 +4.9	44.0 +4.5
R1-Distill-7B	56.7	40.5			
+ Direct-OPD	63.1 +6.4	48.8 +8.3			

Table 1 Score summaries for the different students and teacher pairs

3.2 Weak-to-strong generalization beats RL

We next ask whether Direct-OPD is only a way to reuse an already trained small teacher, or whether it can be a better training path than running RL directly on the larger target model. We compare two matched-step routes. The direct-RL route trains R1-Distill-7B with RL. The weak-to-strong route first trains the smaller R1-Distill-1.5B model with RL, then transfers the resulting RL/reference policy shift into R1-Distill-7B using Direct-OPD.

We test that for the same number of RL steps, a smaller model is a more efficient place to find a useful policy-improvement direction, and Direct-OPD can then transfer that direction to the larger student. We trained R1-Distill-1.5B on DAPO dataset, and select R1-Distill-1.5B RL checkpoints at steps 300, 600, 900, 1200, 1500. For each selected checkpoint, we construct a teacher pair using the base R1-Distill-1.5B model as $\pi_{T_{\text{ref}}}$ and the selected RL checkpoint as π_T . We then train R1-Distill-7B with Direct-OPD and compare the resulting validation score against direct R1-Distill-7B RL at the matched small-teacher RL step. In the figure, T300 denotes the route that first trains R1-Distill-1.5B with RL for 300 steps and then applies Direct-OPD to R1-Distill-7B; T600–T1500 are defined analogously.

We find that, for the same number of RL steps, running RL on the small model and then transferring the learned policy shift to the large model gives better R1-Distill-7B performance than running RL directly on R1-Distill-7B.

This matched-step advantage also translates into a compute advantage. In our setup, a 1500-step RL run on R1-Distill-1.5B takes about 160 hours on 32 A100 GPUs, while RL on R1-Distill-7B takes about 320 hours. After small-model RL, the Direct-OPD transfer stage adds only about 4 hours on 8 A100 GPUs, which is negligible compared with the RL cost. This low transfer cost comes from two factors: Direct-OPD requires only a short training run, as discussed in Section 4.2, and the teacher models used for scoring are small enough that scoring is not the speed bottleneck.

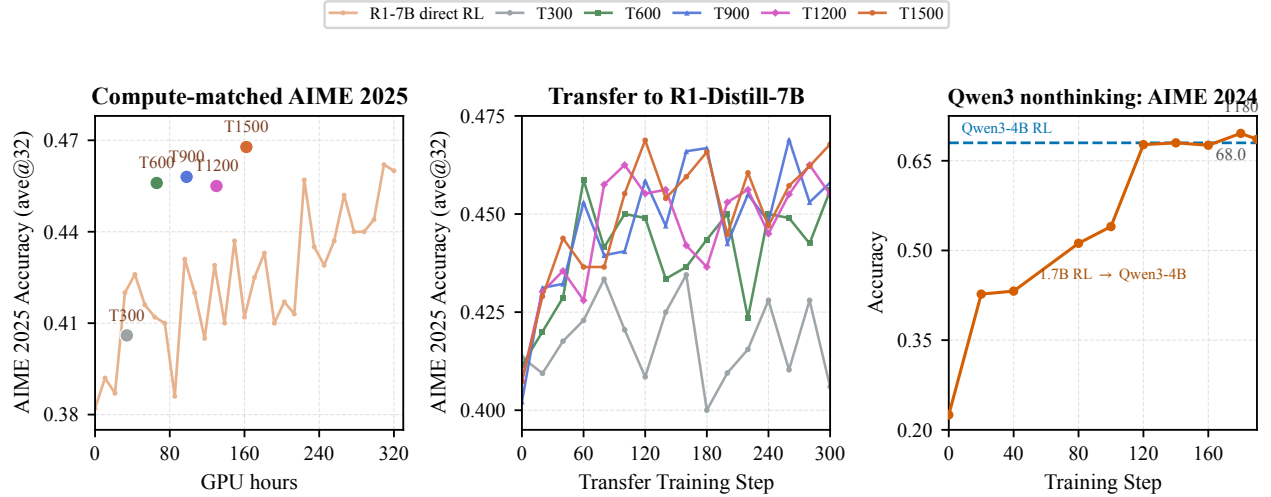


Figure 3 Running RL on a small model and transferring its policy shift with Direct-OPD beats running RL directly on the large target, at equal compute. We compare two routes to improving R1-Distill-7B: direct RL on R1-Distill-7B, versus a weak-to-strong route that runs RL on the smaller R1-Distill-1.5B and transfers the resulting policy shift into R1-Distill-7B with Direct-OPD. T_N denotes the transfer that uses the R1-Distill-1.5B RL checkpoint at step N as the post-RL teacher π_T , with the base model as $\pi_{T_{\text{ref}}}$. **Left:** AIME 2025 accuracy against total GPU-hours; the wiggly curve is direct R1-Distill-7B RL, and each T_N point sums its small-model RL cost and the short Direct-OPD transfer. Later transfers (T600–T1500) sit above the direct-RL curve at equal compute—higher accuracy for less compute. **Middle:** Direct-OPD transfer trajectories into R1-Distill-7B from the five small-teacher checkpoints; the early T300 carries a weaker shift than T900–T1500. **Right:** the same recipe with Qwen3 non-thinking models—transferring a Qwen3-1.7B RL shift into Qwen3-4B reaches the 68.0 accuracy of direct Qwen3-4B RL (dashed) on AIME 2024.

Figure 3 shows that the transferred signal depends on which small-teacher RL checkpoint is used. This dependence is expected: different points in the small-model RL trajectory encode different policy shifts, and not every shift is equally useful on the larger student’s state distribution.

This result changes the role of the small model. The small teacher is not used because its endpoint policy is stronger than R1-Distill-7B. Instead, the small model provides a cheaper environment in which RL discovers a direction of improvement. Direct-OPD then transfers that direction to the larger model by evaluating the teacher/reference log-ratio on the larger student’s own on-policy states. Direct RL on the large model must discover the same kind of direction through its own rollouts; weak-to-strong transfer reuses the direction already found by the small model.

We observe the same pattern with Qwen3 nonthinking models. After a 100-step RL run on the 1.7B model, transferring the resulting policy shift to Qwen3-4B-nonthinking reaches the 68.0 direct-RL level on AIME 2024. This suggests that a small-model RL run can recover the useful RL direction even when the target is a stronger 4B model.

Takeaway. Compared with running RL directly on the large model, first running RL on the small model and then transferring to the large model has two advantages: small-model RL is more stable, easier, and cheaper, while the following Direct-OPD stage provides fast transfer to the large model while preserving the performance advantage.

3.3 Sequential composition

This experiment asks whether two independently learned policy shifts can be applied in sequence to the same student. The student is first trained with the R1-Distill-1.5B $\rightarrow$ JustRL-1.5B signal and then continued with the Nemotron-1.5B $\rightarrow$ QuestA-Nemotron-1.5B signal. Figure 4 shows the AIME 2024 trajectory after aligning the second stage’s local 0–300 training steps to global steps 300–600.

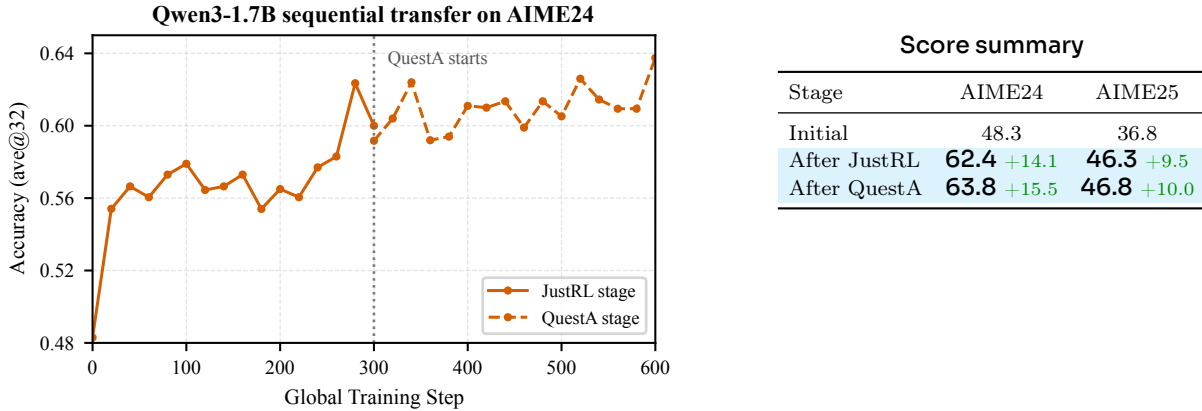


Figure 4 Sequential policy-shift transfer into Qwen3-1.7B on AIME 2024. Left: the AIME 2024 trajectory after aligning the second stage to global steps 300–600. Right: endpoint scores on AIME 2024/2025. The first stage uses the R1-Distill-1.5B $\rightarrow$ JustRL-1.5B signal; the second stage continues from that checkpoint with the Nemotron-1.5B $\rightarrow$ QuestA-Nemotron-1.5B signal.

Takeaway. Different RL training runs can learn different abilities, and Direct-OPD can sequentially compose these abilities into the same student.

4 Analysis and Training Dynamics

The previous section showed that Direct-OPD transfers RL gains weak-to-strong; here we analyze when and why it works, examining three aspects of the transfer:

- **What is transferred?** Whether Direct-OPD needs the student to imitate the teacher’s high-probability tokens, or whether it transfers even when teacher–student top- k overlap stays low (Section 4.1).
- **Does short-horizon training generalize?** Whether training on short rollouts changes only the supervised prefixes or shifts behavior across much longer responses (Section 4.2).
- **Why adapt the KL coefficient?** How the KL coefficient keeps the teacher-shift reward reliable across teacher–student pairs (Section 4.3).

4.1 Cross-pattern transfer without token-overlap imitation

Prior work on standard OPD argues that teacher-student thinking-pattern compatibility is a key success condition: successful OPD is reflected in progressive alignment on high-probability tokens at student-visited states, with a small shared top- k token set carrying most of the probability mass [7]. This requirement is

natural for raw teacher imitation, because the student is asked to match the teacher’s final distribution. In that setting, increasing top- k overlap is both a diagnostic and a plausible carrier of the training signal.

We compute top- k overlap using the same set-overlap metric as prior OPD work. For a student policy π_S and a comparison policy π_C at state s_t , let

$$T_k^S(s_t) = \text{TopK}_v \pi_S(v | s_t), \quad (17)$$

$$T_k^C(s_t) = \text{TopK}_v \pi_C(v | s_t). \quad (18)$$

The per-state overlap ratio is

$$\text{Overlap}_k(S, C; s_t) = \frac{|T_k^S(s_t) \cap T_k^C(s_t)|}{k}. \quad (19)$$

For Direct-OPD, we report this metric against both the post-RL teacher ($C = T$) and the teacher reference ($C = T_{\text{ref}}$) on the same student-visited states.

Direct-OPD changes the object being transferred. Instead of matching the post-RL teacher distribution, it evaluates the teacher/reference log-ratio on student-visited candidate tokens. This signal can rank actions within the student’s own support, so useful transfer need not require the student to enter the teacher’s high-overlap token regime.

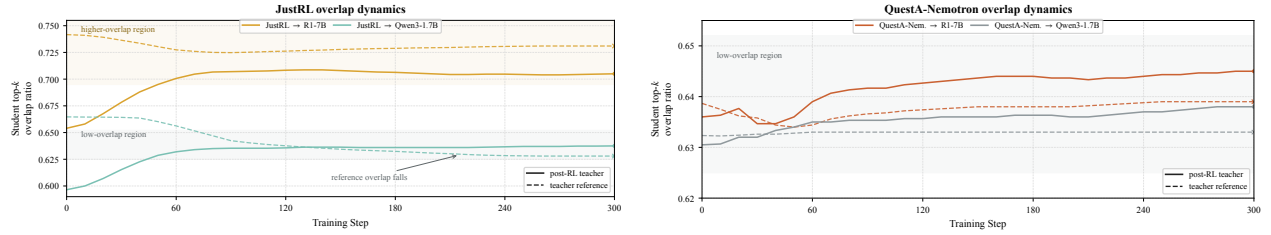


Figure 5 Teacher/student top- k overlap during Direct-OPD training. Left: JustRL teacher pairs. Right: QuestA-Nemotron teacher pairs. Solid curves measure overlap with the post-RL teacher, and dashed curves measure overlap with the teacher reference. The pattern-aligned JustRL into R1-Distill-7B setting enters a higher-overlap regime, while the cross-pattern transfers remain lower and do not become endpoint imitation.

Figure 5 separates the pattern-aligned case from the cross-pattern cases. The aligned transfer follows the standard OPD intuition: the student moves into a higher-overlap regime with the post-RL teacher. The cross-pattern transfers behave differently. Their overlap with the post-RL teacher remains lower, and their overlap with the teacher reference does not show a compensating rise. Thus the gains in Figure 2 are not explained by progressive imitation of either endpoint policy. They are better explained by using the RL-induced direction from reference to post-RL teacher on the student’s own visited states.

We next check whether this low-overlap transfer is a degenerate sharpening effect. If Direct-OPD simply collapsed the actor or forced a trivial low-entropy distribution, the overlap diagnostic would be less informative.

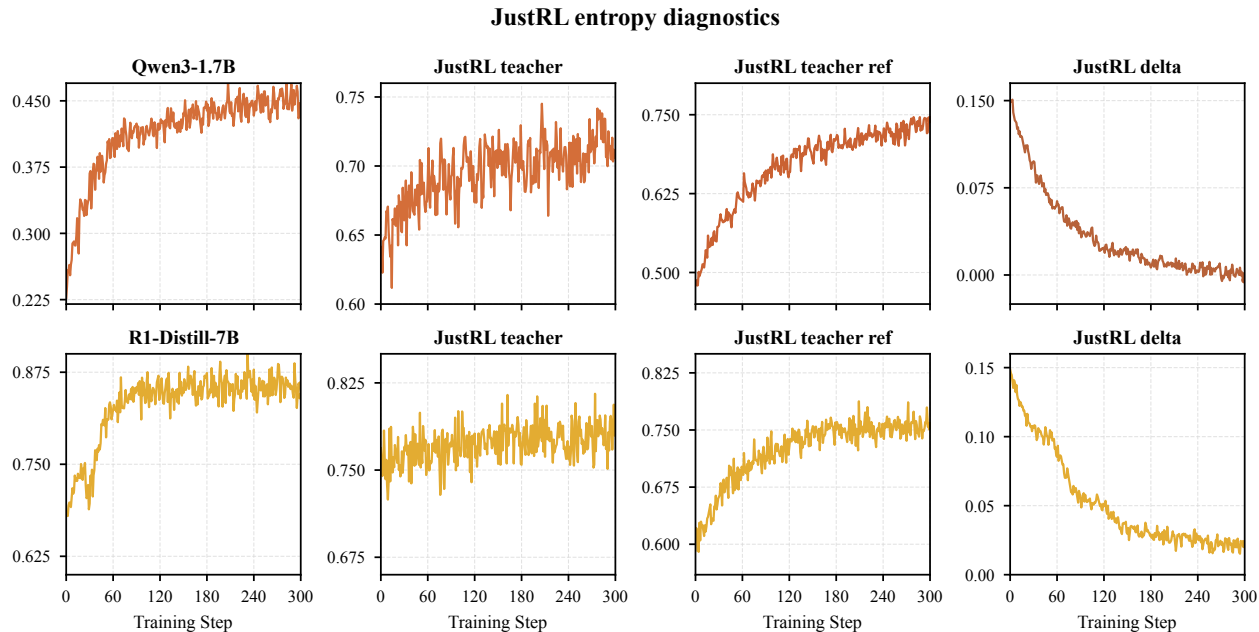


Figure 6 Entropy diagnostics for JustRL policy-shift transfer. **Top row:** JustRL into Qwen3-1.7B. **Bottom row:** JustRL into R1-Distill-7B. Each row shows student entropy, post-RL teacher entropy, teacher-reference entropy, and teacher entropy minus reference entropy. Actor entropy does not collapse, while the teacher/reference entropy gap narrows over training.

Figure 6 shows that the actor entropy remains controlled rather than collapsing. At the same time, the teacher/reference entropy gap narrows, indicating that training changes how the student samples regions where the teacher’s RL-induced shift is evaluated. This supports the same mechanism as the overlap result: Direct-OPD transfers a local policy-shift signal within the student’s distribution, not a full endpoint policy.

Takeaway. Direct-OPD does not require progressive token-overlap imitation. It can transfer the teacher’s RL-induced direction on student-visited tokens while keeping the actor distribution controlled.

4.2 Short-horizon training changes longer-rollout behavior

Recent work on on-policy prefix distillation shows that full-length supervision is not always necessary for reasoning transfer: student rollouts can be truncated to short prefixes, while the resulting model is still evaluated with long generations at test time [9]. Following this view, we do not treat the training horizon merely as a finite rollout length. We deliberately use a short 2k-token response horizon for Direct-OPD training, and ask whether the resulting policy change generalizes beyond the directly supervised prefix.

A natural concern is that such short-horizon training may only change the early prefixes that receive direct supervision, leaving later reasoning behavior unchanged. We test this by evaluating trained actors on fixed long rollouts and measuring whether the actor’s likely next tokens move toward the post-RL teacher relative to the teacher reference.

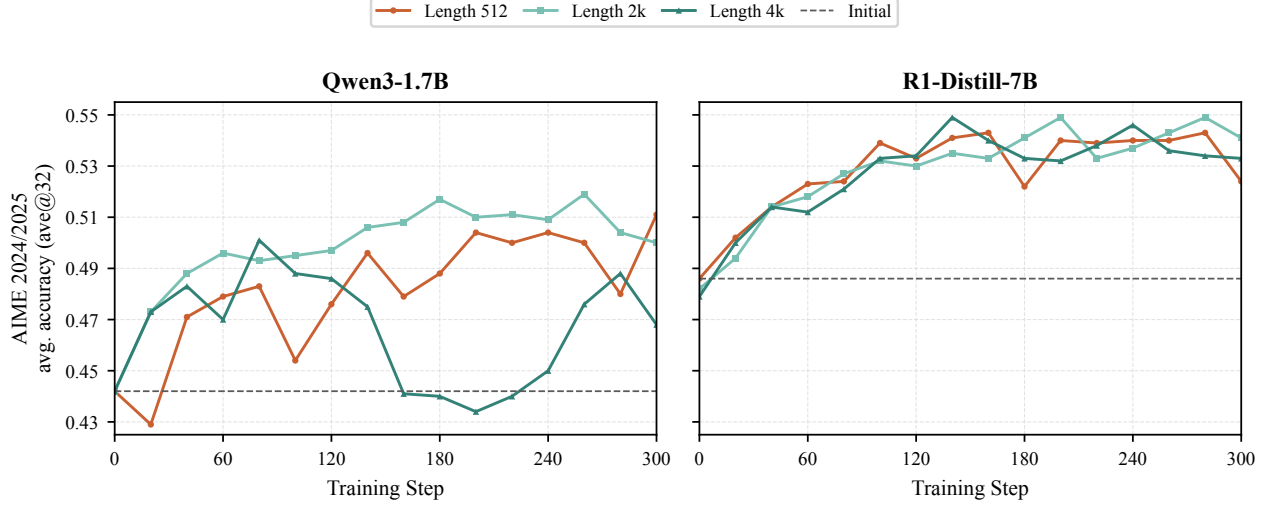


Figure 7 JustRL response-length sweeps with fixed KL coefficient 1. We report the average of AIME 2024 and AIME 2025 validation accuracy (ave@32) during training for Qwen3-1.7B and R1-Distill-7B students. The 2k setting gives stable validation behavior across the two students, while shorter or longer rollouts do not consistently improve the validation curves.

For each response prefix $x_{\leq t}$, we take the actor’s top-16 token set K_t and compute the actor-probability-weighted teacher gap

$$g_t = \sum_{a \in K_t} \pi_{\text{actor}}(a \mid x_{\leq t}) [\log \pi_{\text{JustRL}}(a \mid x_{\leq t}) - \log \pi_{\text{R1}}(a \mid x_{\leq t})]. \quad (20)$$

The plotted quantity is the prefix cumulative gap

$$G_T = \sum_{t=1}^T g_t. \quad (21)$$

Larger G_T means that, along the actor’s own likely tokens, the JustRL teacher assigns higher weighted log-probability than the R1-Distill-1.5B teacher; smaller G_T means that the actor remains closer to R1-Distill-1.5B.

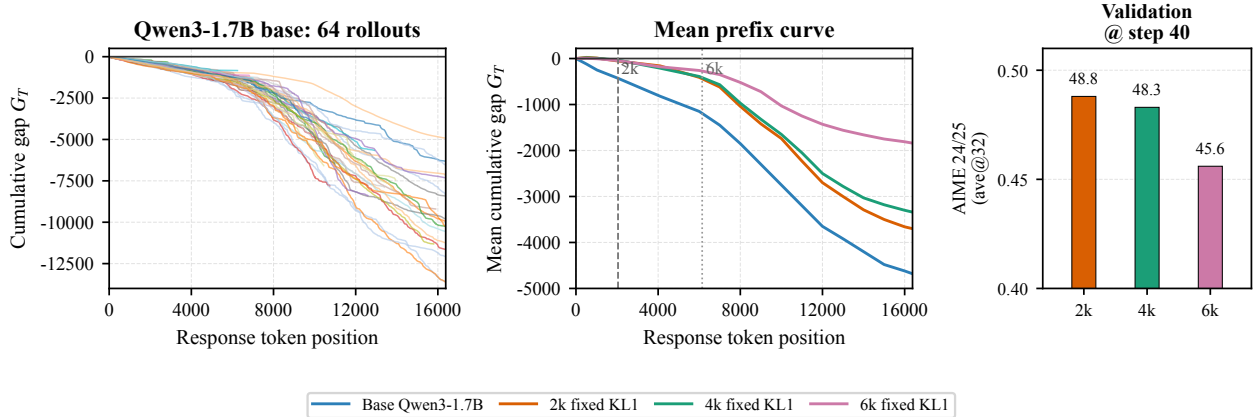


Figure 8 Training Direct-OPD on short rollouts shifts the actor toward the teacher direction far beyond the trained length. On a fixed set of 64 long rollouts we track the cumulative gap $G_T = \sum_{t \leq T} g_t$, where g_t weights, over the actor’s top-16 tokens at position t , the log-probability the post-RL teacher (JustRL) assigns minus that of the reference (R1-Distill-1.5B); higher G_T means the actor’s likely tokens look more like the post-RL teacher, while lower means the actor remains closer to the reference. **Left:** the 64 per-rollout trajectories for the untrained Qwen3-1.7B actor all drift negative—its long rollouts are reference-like, and this is not driven by a single outlier. **Middle:** the mean G_T for the base actor and for actors trained with 2k/4k/6k response length (40 steps, fixed KL=1); every trained actor sits above the base across the full $\sim 16k$ positions, and the 2k actor shifts well past its 2k training horizon (dashed line). **Right:** AIME 2024/2025 (ave@32) at the same 40-step checkpoint—the 2k setting validates best.

Figure 8 shows that the effect of short Direct-OPD training is not confined to the supervised prefix range: after 40 training steps with a 2k response length, the actor moves toward the teacher-shift direction across much longer fixed rollouts, so the short horizon already captures the transferable signal. Training with a 6k response length shifts the actor even further in this diagnostic (middle panel), yet it validates worst (right panel, 45.6 versus 48.8 for 2k). The reason is *where* that extra movement comes from: the late prefixes, where the small teacher pair is increasingly off-distribution and the teacher/reference log-ratio is large but unreliable (Section 4.3). A 6k horizon over-drives the actor on this unreliable late-prefix signal without improving the answer, whereas a 2k horizon avoids ingesting it while still capturing—and generalizing—the reliable early-prefix direction.

Takeaway. Short-horizon Direct-OPD training can generalize changes in thinking patterns to longer rollouts. A moderate response length avoids the noise and instability of very long responses while capturing more useful signal than overly short training horizons.

4.3 KL keeps the teacher-shift reward reliable

The teacher/reference log-ratio acts as a dense reward, but this reward is only reliable on states where the teacher shift is meaningful for the student. This makes KL more than a regularization coefficient: it controls which rollout states the student visits, and therefore whether the dense reward remains reliable. If KL is too weak, the student can drift into regions where both teacher and reference assign low probability and the log-ratio becomes a noisy training target. If KL is too strong, the student cannot follow the teacher’s policy shift. We test this by sweeping fixed KL coefficients under the same 2k-response setting and comparing them with an adaptive-KL run.

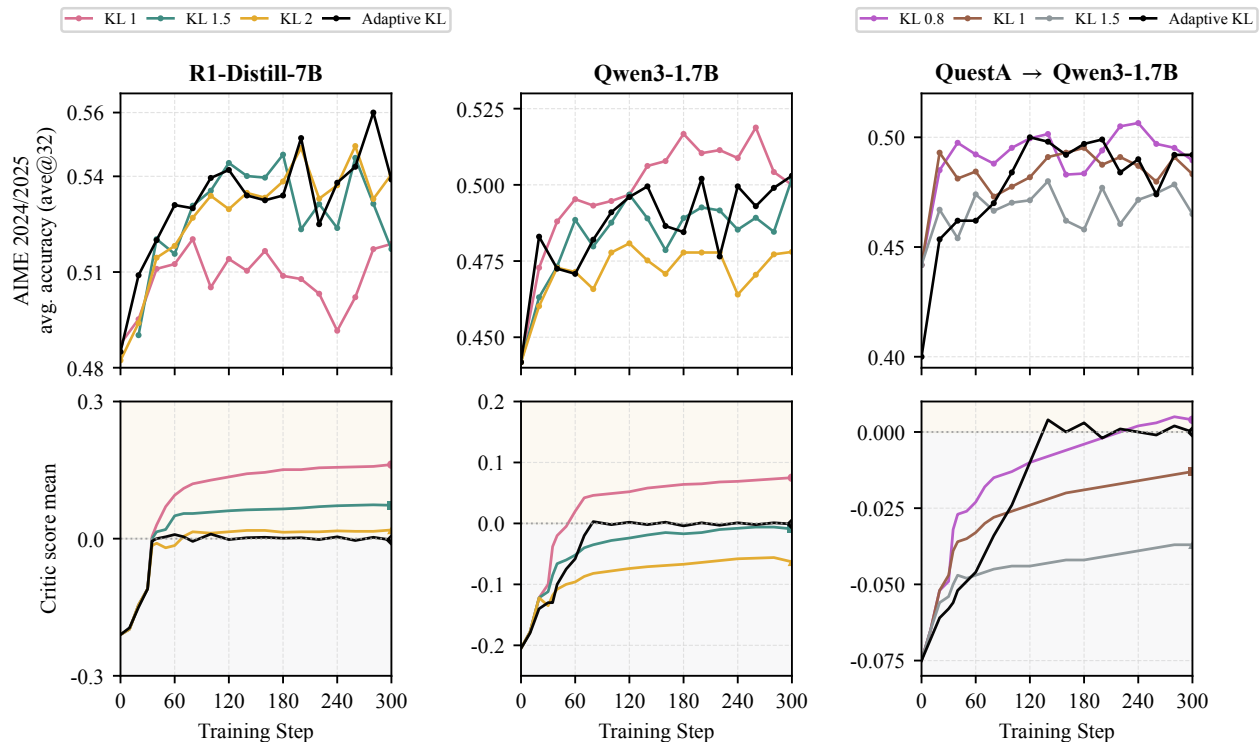


Figure 9 The best KL is pair-dependent, and adaptive KL keeps the dense reward in its informative regime near zero. 2k-response runs across fixed KL coefficients, with adaptive KL as the black curve. **Top row:** AIME 2024/2025 validation accuracy (ave@32); **bottom row:** mean critic score (the dense teacher/reference reward). The best fixed coefficient differs across teacher-student pairs, and a larger critic score does not imply better validation. Adaptive KL instead drives the critic score toward zero after an initial correction, keeping the student where the teacher/reference comparison stays informative rather than maximizing the reward.

Figure 9 shows that the best fixed KL value is not universal: different teacher-student pairs prefer different constraint strengths. The critic score also does not provide a pair-independent rule; a larger positive reward can coincide with worse validation in one setting but track the best validation curve in another. Thus the dense reward should not be treated as a scalar objective to maximize independently of the rollout distribution. The useful regime depends on the teacher-student pair and on where the student currently samples.

The adaptive-KL runs provide a complementary diagnostic. After an initial correction phase, their critic rewards are pulled toward zero rather than remaining strongly positive or negative. This behavior matches the intended role of adaptive KL: keep the student in a region where the teacher/reference comparison is informative, instead of letting the student either ignore the teacher shift or exploit an unreliable local signal. Intuitively, a persistently large reward means the student is sampling tokens on which the post-RL teacher and its reference strongly disagree—typically because it has drifted away from both models’ support, where the log-ratio is large in magnitude but no longer a trustworthy improvement direction. A reward near zero instead keeps the student where the two teachers are close to balanced on average, so the local per-token shifts (some positive, some negative) still rank actions meaningfully without a global drift; this balanced regime is where the signal stays *reliable*.

Takeaway. The dense teacher/reference reward must be kept reliable rather than maximized. Fixed-KL sweeps show that the right constraint is pair-dependent, while adaptive KL keeps the reward near a balanced regime, preventing the student from either ignoring the teacher shift or exploiting unreliable rollout states.

5 Related Work

On-policy distillation of reasoning models. Reinforcement learning with verifiable rewards now produces strong reasoning models [1–4, 6, 8, 10–16], a paradigm now standard in frontier systems [17–22]. A standard way to spread such gains to other models is knowledge distillation, from classic and sequence-level distillation to its modern variants and scaling behavior [23–30], with a large body of work on divergences and estimators for distilling LLMs [31–39]. On-policy distillation (OPD), where the student is trained on its own sampled states under teacher supervision, has become the dominant variant for reasoning models and is the subject of rapid recent work [7, 40–60], including hybrids that couple distillation with RL [61–66]. Analyses tie OPD success to teacher–student top- k overlap on student-visited states [7], note that small students struggle to copy much stronger reasoners [67], and reinterpret OPD as dense KL-constrained RL that rescales the reward to extrapolate beyond the teacher [68]. All of these imitate, or push past, the teacher’s endpoint policy; when the teacher is a small post-RL model weaker than the target, this also imports its capacity ceiling. Direct-OPD instead distills only the teacher’s RL-induced policy shift $\log \pi_T - \log \pi_{T_{\text{ref}}}$, discarding the teacher’s absolute policy.

Weak-to-strong generalization. Weak-to-strong generalization asks whether a weak supervisor can elicit capabilities from a stronger model [69, 70], building on a long line of learning from imperfect supervision—semi-supervised learning and learning from noisy labels [71–80]. It is central to scalable oversight when reliable human or stronger-model supervision is scarce [81–86], and is related to eliciting latent knowledge and to easy-to-hard generalization [87–90]. Recent work studies its theory and extends it to reasoning and automated alignment [91–95]. Most approaches still supervise with the weak model’s labels or distribution, which caps the student near the supervisor; self-rewarding schemes bootstrap a model with its own judgments [96], and, closest to us, weak-to-strong preference optimization reuses a weak aligned model’s implicit reward to steer a stronger one [97]. We share the goal of eliciting rather than imitating, but our supervisor is a pair of small RL checkpoints: we transfer the reward implied by their difference, and show it improves students that already exceed the post-RL teacher.

Implicit rewards and reusing trained models. Direct-OPD rests on the policy-as-reward identity: under KL-regularized RL the policy/reference log-ratio recovers the reward up to a constant, which Direct Preference Optimization and its variants use to fit a policy directly from preferences without an explicit reward model [5, 98–100]. We use this identity in reverse—reading a dense implicit reward off a post-RL checkpoint and its reference—which connects to dense and process rewards for reasoning [101–103] and to per-token credit assignment in RL [104, 105], while avoiding reward over-optimization from an explicit model [106]. A complementary line reuses trained models in weight space—task arithmetic, model merging, and proxy tuning [107–109]. Direct-OPD differs on both counts: the signal is a behavioral log-ratio rather than a weight delta, and it is applied on the student’s own on-policy states, so it transfers across model families and scales.

6 Conclusion and Limitations

Direct-OPD shows that the policy shift learned by a small RL teacher is a useful dense reward for weak-to-strong generalization. The transferable object is not the post-RL teacher’s policy, but its log-ratio against its own pre-RL reference, evaluated on student-visited tokens; this is why a smaller, weaker teacher can still improve stronger students. Empirically, this direction transfers across teacher pairs and student families, beats step-matched direct RL at a fraction of the compute, and composes across teachers. It reframes RL outcomes as reusable improvement signals rather than final models to imitate. **Limitations.** The signal is conditional: Direct-OPD can fail when the teacher/reference improvement is not meaningful on student-visited states, and the best response length and KL strength remain teacher–student dependent.

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Experimental Details

All Direct-OPD experiments use the student’s on-policy rollouts and evaluate the teacher/reference log-ratio on the student’s top- k candidate tokens. Unless a section states otherwise, reported validation curves use AIME 2024 and AIME 2025 with the same evaluation protocol within each comparison, so the main comparisons isolate the transferred policy shift, teacher pair, response length, or KL coefficient rather than the benchmark protocol.

Teacher pairs. The main experiments use two teacher/reference pairs: R1-Distill-1.5B $\rightarrow$ JustRL-1.5B and Nemotron-1.5B $\rightarrow$ QuestA-Nemotron-1.5B. In each case, Direct-OPD transfers the policy shift from the reference model to the post-RL teacher rather than the endpoint teacher distribution.

Direct-OPD data and prompt. For the final Direct-OPD runs, we use the math subset of Skywork-OR1-RL-Data, released with Skywork-OR1 [110], and apply the following DAPO-style math prompt template.

Solve the following math problem step by step.
 The last line of your response should be of the form
 Answer: \$Answer (without quotes) where \$Answer is the answer to the problem.

{Question}

Remember to put your answer on its own line after "Answer:".

We use this template for Direct-OPD training rollouts and evaluation prompts. This template differs from the boxed-answer prompt used during teacher RL; in our runs, the DAPO-style prompt gives slightly better transfer. Since prompt template divergence is not the focus of this paper, we use this setting throughout. We also observe similar transfer trends when replacing the Skywork training dataset with DAPO-Math-17K while keeping the same prompt format, suggesting that the result is not specific to a single training dataset.

Evaluation. Table 2 summarizes the evaluation protocol used for all reported AIME results unless otherwise specified.

Evaluation setting	Value
Benchmarks	AIME 2024, AIME 2025
Samples per problem	32
Sampling temperature	0.7
Top- p	0.95
Maximum generation length	31,744

Table 2 Evaluation protocol for AIME validation.

Table 3 lists the default Direct-OPD training hyperparameters used with the data and prompt format above.

Hyperparameter	Value
Training framework	verl
Global batch size	64
Mini batch size	64
Rollout n	4
Maximum prompt length	1,024
Maximum response length	2,048
Sampling temperature	1.0
Top- p	1.0
Learning rate	1×10^{-6}
Training epochs	1
KL coefficient	0.0 unless otherwise specified
Student top- k support	16
Top- k strategy	Student top- k
Adaptive KL controller	Used in adaptive-KL experiments only

Table 3 Training hyperparameters for Direct-OPD unless otherwise specified.

Table 4 reports the GRPO settings used to train RL teachers and direct-RL baselines.

R1-Distill-1.5B RL		R1-Distill-7B RL	
Hyperparameter	Value	Hyperparameter	Value
Algorithm	GRPO	Algorithm	GRPO
Train batch size	512	Train batch size	512
PPO mini batch size	128	PPO mini batch size	128
Rollout n	8	Rollout n	8
Maximum prompt length	2,048	Maximum prompt length	2,048
Maximum response length	16,384	Maximum response length	16,384
Temperature	1.0	Temperature	1.0
Learning rate	1×10^{-6}	Learning rate	1×10^{-6}
KL coefficient	0.0	KL coefficient	0.0
Clip high	0.28	Clip high	0.28
Clip low	0.2	Clip low	0.2

Qwen3-1.7B-Nonthinking RL		Qwen3-4B-Nonthinking RL	
Hyperparameter	Value	Hyperparameter	Value
Algorithm	GRPO	Algorithm	GRPO
Train batch size	128	Train batch size	128
PPO mini batch size	128	PPO mini batch size	128
Rollout n	8	Rollout n	8
Maximum prompt length	2,048	Maximum prompt length	2,048
Maximum response length	16,384	Maximum response length	16,384
Temperature	1.0	Temperature	1.0
Learning rate	1×10^{-6}	Learning rate	1×10^{-6}
KL coefficient	0.0	KL coefficient	0.0

Table 4 RL training settings for teacher construction and direct-RL baselines. We follow the math GRPO setting of Yang et al. [68], using train batch size 512 for R1-Distill runs, train batch size 128 for Qwen3-Nonthinking runs, and PPO mini batch size 128 throughout.

Additional Entropy Diagnostics

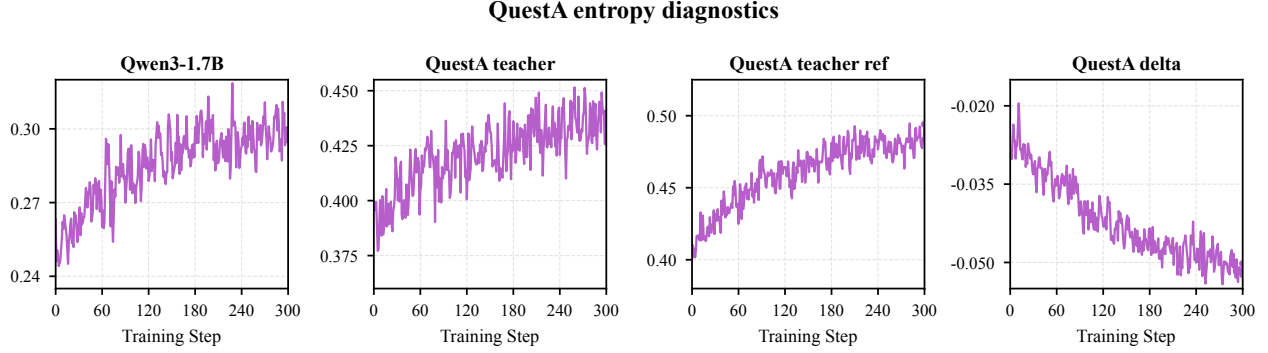


Figure 10 Entropy diagnostics for QuestA-Nemotron into Qwen3-1.7B. The panels show student entropy, post-RL teacher entropy, teacher-reference entropy, and teacher entropy minus reference entropy. Together with Figure 6, this shows that the non-collapse pattern is not specific to the JustRL teacher pair.

Additional Sequential Composition Figures

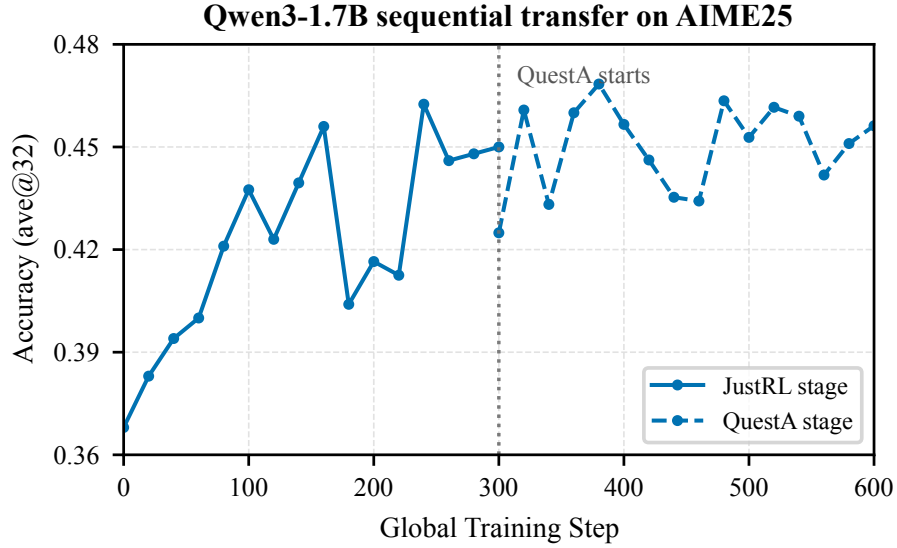


Figure 11 Sequential policy-shift transfer into Qwen3-1.7B on AIME 2025. The QuestA-Nemotron stage is aligned after the JustRL stage using global steps 300–600, matching the AIME 2024 alignment in Figure 4.

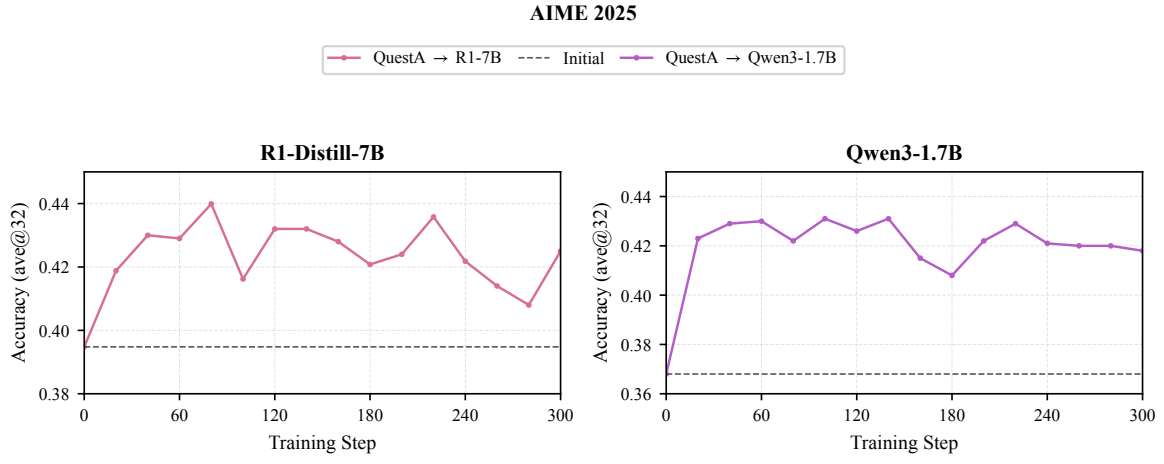


Figure 12 QuestA transfer curves on AIME 2025 for the cross-pattern transfer setting. The main text reports the corresponding AIME 2024 curves.